



CII RISK FORECAST REPORT

MT CASPIAN VOYAGER

IMO 9841036

MR2 Product Tanker | DWT: 51,200 | Built: 2018

Flag: Marshall Islands | Class: DNV | Fuel: VLSFO (CF 3.114)

FORECAST PERIOD

2026-2030

Prepared for: Oceanus Maritime Ltd.

Ref: NTX-2025-08472

VESSEL PARTICULARS

Vessel Name	IMO Number
MT CASPIAN VOYAGER	9841036
Type	DWT
MR2 Product Tanker	51,200
Built	Fuel
2018	VLSFO (CF 3.114)
Flag	Class
Marshall Islands	DNV

MACHINERY

Main Engine	MCR
MAN B&W; 6S50ME	8,900 kW
Aux Engine	EPL Status
3x Yanmar 680 kW	ACTIVE
MCR Limit	Manager
75% (6,675 kW)	Oceanus Maritime

REPORT DETAILS

Prepared For	Reference
Oceanus Maritime Ltd.	NTX-2025-08472
Period	
2026-2030	

CII OVERVIEW

Attained CII	Required CII
4.99	4.71
CII Rating	Reduction Factor
C — MEDIUM	11%

C

MEDIUM

AI RISK ASSESSMENT

OVERALL RISK: HIGH

Marginal C in 2026 -> D from 2027 -> CAP trigger by 2029

01 OPERATIONAL BASELINE — ANNUAL PROFILE

FUEL (MT)	DIST. (NM)	SEA DAYS	AVG SPD	CO2 (T)	ATT. CII	FUEL TYPE	AER UNIT
6,480	79,000	250	13.2 kn	20,179	4.99	VLSFO	gCO2/DWT·NM

$AER = (6,480 \times 3.114 \times 10^6) / (51,200 \times 79,000) = 4.99$. Emission factor 3.114 t CO2/t VLSFO per MEPC.364(79).

02 VOYAGE EFFICIENCY BREAKDOWN

SEGMENT	% DISTANCE	DISTANCE (NM)	AVG SPEED	FC/DAY (MT)	SEGMENT FC
LADEN	55%	43,450	12.5 kn	27.5	3,988 MT
BALLAST	45%	35,550	14.0 kn	22.0	2,321 MT
PORT/IDLE	---	---	---	4.2	483 MT

Laden: Med/NW Europe CPP trade. Ballast: return positioning. Port ~90 days/yr. Idle ~25 days.

03 EEXI COMPLIANCE STATUS

REQUIRED EEXI	ATTAINED EEXI	COMPLIANCE	EPL STATUS	MCR LIMITED	IEE CERT
5.92	5.28	COMPLIANT	ACTIVE	75% (6,675 kW)	JUN 2023

EEXI per MEPC.333(76). Tanker refline: $1142.7 \times DWT^{(-0.488)}$, 20% reduction. EPL overridable per MEPC.335(76).

04 CII TRAJECTORY FORECAST — 2026-2030

YEAR	Z %	REQ. CII	ATT. CII	C/D BOUND	D/E BOUND	RATING	RISK
2026	11%	4.71	4.99	5.09	6.03	C	MEDIUM
2027	13%	4.60	4.99	4.97	5.89	D	HIGH
2028	15%	4.50	4.99	4.86	5.76	D	HIGH
2029	17%	4.39	4.99	4.74	5.62	D	CRITICAL
2030	19%	4.28	4.99	4.62	5.48	D	CRITICAL

Constant ops, no mitigation. $CII_{ref} = 5247.2 \times 51200^{(-0.610)} = 5.29$. Boundaries per MEPC.354(78) tanker dd vectors.

05 EU ETS CARBON COST EXPOSURE — 2026-2028

YEAR	TOTAL CO2	EU-APPLIC.	PHASE-IN	SURRENDER	EUA €/T	COST (€)
2026	20,179 T	11,098 T	100%	11,098 T	€75	€832,350
2027	20,179 T	11,098 T	100%	11,098 T	€80	€887,840
2028	20,179 T	11,098 T	100%	11,098 T	€85	€943,330

Route: 30% EU>EU (100%) + 25% EU>non-EU (50%) + 25% non-EU>EU (50%) + 20% non-EU (0%) = 55% effective. Cumulative 3-yr: €2,663,520.

06 FUELEU MARITIME — GHG INTENSITY & PENALTY

BASELINE	REDUCTION	GHG LIMIT	VESSEL GHG	DEFICIT	PENALTY/YR
91.16 gCO ₂ eq/MJ	2% (2025-29)	89.34	91.74	+2.40	€227,000

VLSFO $WtW = 91.74$ gCO₂eq/MJ (FuelEU Annex II). Exceeds limit by 2.7%. Options: B20 biofuel, LNG pooling, RFNBO (2x reward).

07 MITIGATION IMPACT SCENARIOS

MEASURE	CII IMPACT	NEW CII	2027 RTG	2030 RTG	EU ETS Δ	FUELEU
NO ACTION (BASELINE)	0%	4.99	D	D	€0	NON COMPLIANT
SPEED 12.0 KN (-6%)	-6%	4.69	C	D	-€50K	NON COMPLIANT
SPEED + HULL CLEAN (-10%)	-10%	4.49	C	C	-€83K	NON COMPLIANT
ABOVE + B20 BIOFUEL	-10%	4.49	C	C	-€83K	COMPLIANT
LNG DUAL-FUEL RETROFIT	-23%	3.84	B	B	-€191K	COMPLIANT

CII impacts cumulative. EU ETS savings at 2027 EUA. FuelEU compliant requires ≤ 89.34 ; B20 $\rightarrow \sim 85.8$ gCO₂eq/MJ.

08 POSEIDON PRINCIPLES & SEA CARGO CHARTER

YEAR	VESSEL AER	PP TRAJECTORY	ALIGNMENT Δ	PP STATUS	SCC
2026	4.99	~ 5.15	-0.16	ALIGNED	ALIGNED
2027	4.99	~ 5.00	-0.01	MARGINAL	MARGINAL
2028	4.99	~ 4.85	+0.14	MISALIGNED	MISALIGNED
2029	4.99	~ 4.70	+0.29	MISALIGNED	MISALIGNED
2030	4.99	~ 4.55	+0.44	MISALIGNED	MISALIGNED

PP v5.0 aligned with IMO 2023 Strategy. Tanker 40-60k DWT class. SCC uses same AER method. Misalignment may trigger 5-15 bps margin adjustment.

09 CARBON COST COMPARISON — EU ETS VS VCM

MECHANISM	PRICE/T CO2	ANNUAL COST	COVERAGE
EU ETS (MANDATORY)	€75-85	€832K-943K	55% OF EMISSIONS
FUELEU PENALTY	~€11.25/t equiv.	~€227K	55% OF ENERGY
VCM OFFSET (VERRA VCS)	\$10-15	\$202K-303K	100% (VOLUNTARY)

EU ETS is a compliance obligation. VCM shown for ESG reporting only. Combined mandatory: ~€1.06M/yr (2027). 3-year: ~€3.34M.

10 PORT STATE CONTROL & COMMERCIAL RISK

ELEVATED RISK | CII D/E ratings increase PSC targeting & reduce chartering eligibility

Major oil companies increasingly require minimum C rating for SIRE/CDI vetting. D-rated vessels face: higher P&I; premiums, exclusion from green corridors, and adverse Poseidon-linked financing terms (5-15 bps). PSC risk-based targeting (Paris/Tokyo MoU) increases inspection frequency for poor CII performers.

11 COMPLIANCE TIMELINE & CAP TRIGGER

DATE	ACTION REQUIRED
31 AUG 2025	Submit revised SEEMP Part III (CII targets 2027-2030) to Administration
31 DEC 2025	Confirmation of Compliance (CoC) for revised SEEMP Part III on board
1 JAN 2026	Begin enhanced DCS data collection per MEPC.401(83)
31 JAN 2026	Submit FuelEU Maritime Report for 2025 to accredited verifier
31 MAR 2026	Submit 2025 EU ETS emissions report via THETIS-MRV
30 APR 2026	FuelEU pooling/banking deadline in FuelEU Maritime database
30 JUN 2026	FuelEU Document of Compliance (DoC) required on board; penalty deadline
2027	First projected D rating - begin CAP development
2029	Third consecutive D year - CAP submission mandatory (MARPOL Annex VI Reg. 28)

Three consecutive D ratings -> mandatory CAP per MARPOL Annex VI Reg. 28. Single E -> immediate CAP.

12 AI RISK ASSESSMENT — SUMMARY

OVERALL RISK: HIGH

Marginal C in 2026 -> D from 2027 -> CAP trigger by 2029

The vessel holds a marginal C rating in 2026 with only 2% headroom. By 2027, tightening Z factors trigger D classification through 2030. Three consecutive D years (2027-2029) mandate CAP submission. FuelEU GHG intensity exceeded by 2.7% incurs ~€227K/yr penalties. Combined EU ETS + FuelEU mandatory costs: ~€3.34M over 2026-2028. Poseidon alignment deteriorates from 2028.

PRIORITY ACTIONS:

- (1) Speed reduction to 12.0 kn
- (2) Hull & propeller cleaning Q2 2026
- (3) B20 biofuel trial for FuelEU
- (4) Fleet pooling for CII/FuelEU
- (5) Begin CAP documentation pre-2027
- (6) LNG dual-fuel feasibility study

DISCLAIMERS: CII per IMO AER (MEPC.352(78)). Boundaries per MEPC.354(78). Z post-2026 est. pending MEPC review. EEXI per MEPC.333(76). EU ETS per Directive 2023/959. FuelEU per Reg. (EU) 2023/1805. PP v5.0 trajectory estimates. VCM per Verra VCS Q1 2025. All AI-generated analyses should be validated by qualified professionals prior to decisions.